Unit III :: Principles of Thermodynamics

PHY0500304: Heat & Thermodynamics

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Syllabus

Unit III: Principles of Thermodynamics

Thermodynamic preliminaries: Extensive and intensive properties, Thermodynamic Variables, Thermodynamic Equilibrium, P-V indicator diagram. Work done in terms of P and V, Zeroth Law of Thermodynamics & Concept of Temperature, Internal energy and First Law of Thermodynamics, Applications of First Law: General Relation between CP and CV. Reversible and Irreversible process with examples. Heat & work, state function, Conversion of heat into work and vice versa, Work Done during Isothermal and Adiabatic Processes, Heat Engines, 2nd Law of Thermodynamics: Kelvin-Planck and Clausius Statements and their Equivalence, Carnot's Cycle, Carnot engine & efficiency. Refrigerator & coefficient of performance, Carnot's Theorem. Applications of Second Law of Thermodynamics: Thermodynamic Scale of Temperature and its Equivalence to Perfect Gas Scale.

1 Thermodynamic preliminaries

Here we briefly define systems and surroundings, state and path functions, and the common variables used to characterise a simple compressible system. These preliminaries also set the sign conventions and notation (differentials versus inexact differentials) used in later derivations.

1.1 Extensive and intensive properties

Extensive properties scale with the size (amount of matter) of the system; intensive properties do not. Examples: volume V, internal energy U, entropy S and mass m are extensive (if you double the system, these double). Temperature T, pressure P and density ρ are intensive (they are the same for two identical subsystems in equilibrium).

For an extensive property, as per the principle of mathematical scaling, we have

$$X(\lambda \text{ system}) = \lambda X(\text{system}), \quad \lambda > 0$$

If x is intensive and X is extensive, then $x = \dfrac{X}{N}$ (per-particle or per-mole quantity) is intensive.

1.2 Thermodynamic variables

Thermodynamic variables (state variables) like P, V, T, n (or N) and chemical potential μ specify the macroscopic state of a system.

Some combinations define derived state functions: internal energy $U(S,V,N)$, enthalpy $H(S,P,N) = U + PV$, Helmholtz free energy $F(T,V,N) = U - TS$, and Gibbs free energy $G(T,P,N) = U - TS + PV$. Each potential is convenient for different experimental constraints.

Differentials of state functions are exact – for example, we write

$$\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}v + \mu\,\mathrm{d}N$$

for a simple compressible system (fundamental thermodynamic relation).

1.3 Thermodynamic equilibrium

Thermodynamic equilibrium means no net macroscopic change occurs with time: the system is mechanically, thermally and chemically balanced with its surroundings. Mechanical equilibrium $\implies$ equal pressures; thermal equilibrium $\implies$ equal temperatures; chemical equilibrium $\implies$ equal chemical potentials for exchanging species. At equilibrium (isolated system) entropy is maximum subject to constraints; for a system at fixed T and P, appropriate free energies are minimised.

1.4 P-V indicator diagram

A P-V diagram (indicator diagram) plots pressure vs volume for a thermodynamic process. Paths on this diagram represent processes; closed loops represent cycles. The algebraic area enclosed by a closed path equals the net work done by the system per cycle (conventionally positive if the system does work on the surroundings during an anticlockwise loop).

1.5 Work done in terms of P and V

Work associated with quasi-static (infinitesimally slow) volume change is

$$\mathrm{d}W = P\,\mathrm{d}V$$

where P is the internal (system) pressure if the process is reversible. More generally, for irreversible expansion against a constant external pressure P_{ext}

$$\mathrm{d}W = P_{\text{ext}}\,\mathrm{d}V$$

Total work between states #1 and #2 is

$$W = \int_{V_1}^{V_2} P\,\mathrm{d}V$$

As per the sign convention:

- $W > 0 \implies$ work done *by* the system
- $W < 0 \implies$ work done *on* the system

1.6 Zeroth law of thermodynamics

The zeroth law formalises the concept of temperature.

It states: if system A is in thermal equilibrium with B, and B is in thermal equilibrium with C, then A and C are in thermal equilibrium.

This transitive property allows the construction of temperature scales and thermometers: a single observable (the thermometer reading) can act as a parameter that is equal for all systems in mutual thermal equilibrium.

1.7 Concept of temperature

Temperature is an intensive variable that quantifies the direction of heat flow: heat flows spontaneously from higher to lower temperature. In thermodynamics the absolute (Kelvin) temperature has an intrinsic definition via entropy S and internal energy U as follows:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}$$

Temperature also appears in the ideal gas law $PV = nRT$ and in the statistical definition $T^{-1} = \frac{\partial S}{\partial U}$, linking thermodynamic and microscopic descriptions.

2 Internal energy

The internal energy U is the total energy contained in a system excluding centre-of-mass kinetic and potential energy due to external fields; it includes microscopic kinetic and potential energies (translation, rotation, vibration, intermolecular). U is an extensive state function: its value depends only on the present state, not on the path used to reach it.

For a simple compressible system the fundamental relation can be written

$$\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V + \mu\,\mathrm{d}N$$

For an ideal (classical) monoatomic gas U depends only on temperature: $U = \frac{3}{2}nRT$. Changes in internal energy can arise from heat transfer and work (first law).

3 First law of thermodynamics

The first law is the statement of conservation of energy applied to thermodynamic systems: energy can be transferred as heat or work and internal energy changes accordingly. For a closed system (fixed mass)

$$\Delta U = Q - W$$

where Q is heat supplied to the system and W is work done by the system. In differential form

$$\mathrm{d}U = \delta Q - \delta W$$

- For reversible PV-work, we have $\delta W = P\,\mathrm{d}V$. Hence

$$\delta Q = \mathrm{d}U + P\,\mathrm{d}V$$

- Q and W are path-dependent (inexact differentials, often written as δQ, δW).
- U is a state function (exact differential, written as $\mathrm{d}U$).
- First law does not constrain the direction of processes.

3.1 Applications of the first law of thermodynamics

3.1.1 General relation between C_P and C_V

Specific heats are defined for closed systems of fixed composition:

$$C_V \equiv \left(\frac{\partial U}{\partial T}\right)_V \quad \text{and} \quad C_P \equiv \left(\frac{\partial H}{\partial T}\right)_P$$

where enthalpy $H \equiv U + PV$. For a general substance the difference $C_P - C_V$ can be written in terms of measurable derivatives. One convenient form is:

$$C_P - C_V = -T\frac{(\partial P/\partial T)_V^2}{(\partial P/\partial V)_T}$$

This identity is derived from total differentials and Maxwell relations; it is valid for any simple compressible substance.

- For an ideal gas, we have

$$P = \frac{RT}{V_m}$$

$$\Longrightarrow \left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V_m} \quad \text{and} \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{V_m^2}$$

Substituting the above gives us

$$C_P - C_V = -T\frac{R^2/V_m^2}{\left(-RT/V_m^2\right)}$$

$$\Longrightarrow \boxed{C_P - C_V = R}$$

which is the *Mayer's relation.*

4 Reversible and irreversible processes

A reversible process is an idealised, quasi-static process that can be reversed by an infinitesimal change in conditions, leaving both system and surroundings unchanged. Real processes are irreversible: they proceed in a preferred direction and produce entropy. Typical sources of irreversibility are finite temperature differences, friction, turbulence, unrestrained expansions and inhomogeneities.

Entropy production ΔS_{prod} is zero for reversible processes and positive for irreversible ones. Hence we have

$$\Delta S_{\mathrm{total}} = \Delta S_{\mathrm{system}} + \Delta S_{\mathrm{surroundings}} \geqslant 0$$

with the equality only for reversible transformations.

5 Heat & work

5.1 State function

A state function depends only on the current thermodynamic state and not on the history of the system (e.g. U, S, T, P, V). In contrast, heat Q and work W are path functions – they depend on how the state change is carried out.

Notationally we use $\mathrm{d}U$ for exact differentials and δQ, δW for inexact differentials to emphasise this distinction.

Heat and work are mutually convertible forms of energy. The first law quantifies conversion: when heat Q flows into a system part of it can increase internal energy, and part can be converted to work. The mechanical equivalent of heat (Joule's constant) historically established their equivalence; modern units use joules for both. In practice the efficiency of conversion is limited by the second law.

- A useful consequence: the integral of $\oint \mathrm{d}U = 0$ around a closed cycle, but $\oint \delta Q \neq 0$ and $\oint \delta W \neq 0$ (their difference over a cycle equals zero by the first law).

5.2 Conversion between heat and work

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5.3 Work done during isothermal and adiabatic processes

- **Isothermal process (ideal gas, reversible):**
  T constant, U depends only on T so $\Delta U = 0$. From first law, we have $Q = W$. Therefore, the work done by the gas between volumes V_1 and V_2 is

$$
\begin{aligned}
W_{\text{isothermal}} &= \int_{V_1}^{V_2} P\,\mathrm{d}V \\
&= \int_{V_1}^{V_2} \left(\frac{nRT}{V}\right)\mathrm{d}V \\
&= nRT \int_{V_1}^{V_2} \frac{\mathrm{d}V}{V} \\
\implies W_{\text{isothermal}} &= nRT \ln\left(\frac{V_2}{V_1}\right)
\end{aligned}
$$

- **Adiabatic (reversible) process (ideal gas):**
  Here, $\delta Q = 0$. For an ideal gas the reversible adiabatic relation is

$$PV^\gamma = \text{constant}, \quad \text{where } \gamma \equiv \frac{C_P}{C_V}$$

  Then the work done between states 1 and 2 is

$$W_{\text{adiabatic}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = n C_V (T_2 - T_1)$$

5.4 Heat engines

A heat engine operates in a cycle: it absorbs heat Q_{in} from a hot reservoir, does work W, and expels heat Q_{out} to a cold reservoir. Total work done per cycle is

$$W = Q_{\text{in}} - Q_{\text{out}}$$

The thermal efficiency η is defined as

$$\eta \equiv \frac{W}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$$

The second law of thermodynamics imposes an upper bound on η – no engine operating between two reservoirs can surpass the Carnot efficiency.

6 Second law of thermodynamics

The second law introduces a direction to thermodynamic processes and establishes the concept of entropy as a measure of irreversibility. It can be stated in several equivalent ways (Clausius, Kelvin-Planck) and implies that perpetual motion machines of the second kind are impossible. The law also provides the basis for defining absolute temperature scales via reversible cycles.

6.1 Kelvin-Planck and Clausius statements

Two standard formulations:

1. **Kelvin-Planck statement:** It is impossible to construct a device that operates in a cycle and converts heat completely into work while exchanging heat with a single reservoir. In other words, no cyclic engine can have $Q_{\text{out}} = 0$ for nonzero Q_{in}.

2. **Clausius statement:** It is impossible to construct a device that operates in a cycle and transfers heat from a colder body to a hotter body without work input. That is, heat cannot spontaneously flow from cold to hot.

These two statements are equivalent: violation of one leads to a violation of the other.

6.2 Carnot's cycle

Carnot's cycle is an ideal reversible cycle consisting of two isothermal and two adiabatic processes, operating between a hot reservoir at temperature T_h and a cold reservoir at T_c. The four reversible steps are:

1. Reversible isothermal expansion at T_h absorbing heat Q_h.

2. Reversible adiabatic expansion lowering temperature from T_h to T_c.

3. Reversible isothermal compression at T_c rejecting heat Q_c.

4. Reversible adiabatic compression raising temperature from T_c to T_h.

For a reversible Carnot engine the entropy changes during the isothermal steps are given as $\Delta S = \dfrac{Q}{T}$ and since the cycle is reversible the net entropy change is zero, we have

$$\frac{Q_h}{T_h} - \frac{Q_c}{T_c} = 0$$
$$\implies \frac{Q_c}{Q_h} = \frac{T_c}{T_h}$$

6.3 Carnot engine & efficiency

Using the entropy relation above and the work relation $W = Q_h - Q_c$, the (maximum) efficiency of a reversible Carnot engine is

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h}$$

This depends only on the reservoir temperatures (in Kelvin). No real engine operating between the same two reservoirs can have greater efficiency than a Carnot engine.

For small temperature differences, we have

$$\eta \approx \eta_{\text{Carnot}}$$

6.4 Refrigerator & coefficient of performance

A refrigerator or heat pump runs the Carnot cycle in reverse: work W is supplied to transfer heat Q_c from the cold reservoir to the hot reservoir. The coefficient of performance (α) for a refrigerator is defined as

$$\alpha \equiv \frac{Q_c}{W}$$

Using the relations $W = Q_h - Q_c$ and $\frac{Q_c}{Q_h} = \frac{T_c}{T_h}$, we obtain

$$\begin{aligned} \alpha &= \frac{Q_c}{Q_h - Q_c} \\ &= \frac{Q_c}{Q_c({}^{Q_h}\!/_{Q_c} - 1)} \\ &= \frac{1}{{}^{T_h}\!/_{T_c} - 1} \\ \implies \alpha &= \frac{T_c}{T_h - T_c} \end{aligned}$$

6.5 Carnot's theorem

Carnot's theorem states two important results:

1. No engine operating between two heat reservoirs can be more efficient than a reversible (Carnot) engine operating between the same reservoirs.
2. All reversible engines operating between the same two reservoirs have the same efficiency (independent of working substance).

Consequences: the Carnot efficiency provides an absolute upper bound on thermal efficiency and motivates the definition of an absolute temperature scale tied to reversible heat exchange.

6.6 Applications of second law of thermodynamics

The second law has wide applications:

- *Entropy as a criterion of spontaneity*: for isolated systems, spontaneous processes increase entropy; equilibrium corresponds to maximum entropy.
- *Limits on engine and refrigeration performance*: design targets must respect Carnot limits.
- *Direction of natural processes*: heat flow, diffusion and chemical reactions' spontaneity relate to entropy and free energies.
- *Engineering cycles and optimisation*: second-law analysis identifies irreversibilities and where efficiency improvements are possible.